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Optimization and Hardware Evaluation of a Modified Reconfigurable Ring Oscillator PUF

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Abstract:

Physical Unclonable Functions (PUFs) are emerging hardware security primitives that generate device-specific secret keys by exploiting inherent manufacturing variations. Among the various PUF architectures, Ring Oscillator PUFs (RO-PUFs) are notable for their simplicity and high performance. However, conventional RO-PUFs suffer from a limited number of challenge–response pairs (CRPs), making them susceptible to machine learning (ML) attacks, and scaling them to handle more CRPs often incurs significant hardware overheads. This paper presents a novel Configurable Delay Unit based Modified Reconfigurable Ring Oscillator PUF (MRROPUF) that integrates Exclusive-OR (XOR) and Exclusive-NOR (XNOR) gates to dynamically configure delay paths using challenge bits. This architecture substantially increases the CRP space and entropy while reducing resource utilization. Additional security is achieved through challenge obfuscation via permutation and enhanced reliability via Temporal Majority Voting (TMV). The proposed design was implemented on Xilinx Virtex-6 and Zynq-7000 FPGAs. The experimental results demonstrated high uniqueness (49.76% and 49.0%), near-ideal uniformity (49.3% and 49.0%), and excellent reliability (99.72% and 99.0%) across the two platforms. Compared to existing Configurable RO-PUF and Reconfigurable RO-PUF designs, the MRROPUF achieves superior efficiency and robustness, making it a strong candidate for secure and lightweight hardware authentication in resource-constrained applications.

Keywords: Ring Oscillator, Delay Unit, Temporal Majority Voting (TMV), Hard Macros, Unit resource Cost

1. Introduction

Physical Unclonable Functions (PUFs) have emerged in the past two decades as a widely studied hardware security primitive, offering lightweight and intrinsic protection against cloning and counterfeiting in integrated circuits (ICs) [3, 4]. PUFs can generate device-unique responses to input challenges by exploiting unavoidable manufacturing process variations, thereby enabling applications such as authentication, key generation, and protection of intellectual property (IP) [5]. Unlike secret keys stored in non-volatile memory (NVM), which are vulnerable to invasive readout and tampering [2], PUF responses are generated on demand and do not require permanent storage.

In a PUF, a one-way mapping from an m-bit challenge *C* to an n-bit response R is implemented;

$$\text{PUF: } C \rightarrow R : P(C) = R \quad (1)$$

where *P* is determined by the physical characteristics of the device. The *uniqueness* of this mapping across devices is measured using the inter-chip Hamming distance, Tf_{inter}, and its

reliability under environmental variations is measured using the intra-chip Hamming distance, H_{intra}.

$$H_{inter} = HD\{P(Ch_1), P(Ch_2)\} \quad (2)$$

$$H_{intra} = HD\{P_1(Ch), P_2(Ch)\} \quad (3)$$

PUFs are typically classified as *weak* or *strong* [6, 7]. Weak PUFs, with a limited number of challenge-response pairs (CRPs), are suitable for applications such as key derivation or random number generation, wherein the responses remain internal to the device. Strong PUFs offer a much larger CRP space, supporting challenge response-based authentication without additional cryptographic processors. Prominent strong **PUF** implementations include arbiter PUFs and bistable ring PUFs [8].

Among strong PUF designs, the Ring Oscillator PUF (RO-PUF) has gained particular attention due to its simple architecture, compatibility with standard digital flows, and good statistical performance. A hybrid logic gate-based configurable RO-PUF architecture was proposed in [33] to eliminate the area and power overhead of multiplexer-based designs, and by using compact delay-configurable units, it improves hardware efficiency while preserving PUF properties such as uniformity, uniqueness, and robustness against process and environmental variations. As an extension of [33], an ultra-lightweight configurable RO-PUF structure was proposed in [34] that significantly enhances the CRP space without incurring a proportional silicon area increase and thus can be used in resource-constrained IoT applications. A more robust configurable RO-PUF architecture was proposed in [35] to improve the reliability of PUF- based key generation in embedded systems under environmental variations through oscillator selection optimization. An efficient configurable RO-based module that can be used as both a PUF and a tr random number generator (TRNG) was proposed in [36], which achieved higher hardware efficiency than traditional methods by exploiting logic block configurability, and the statistical evaluation demonstrated strong reliability and uniqueness for the use of such a PUF in embedded and IoT applications. A crossover ring oscillator PUF architecture was proposed in [37] to resist machine learning and differential power analysis attacks, and the proposed architecture increased the CRP space through intercrossed oscillator paths and optimal key selection, thus providing strong attack resilience with low hardware overhead. In [38], a reliability-aware complementary cascode current mirror PUF is presented in 65 nm technology, which, through circuit-level optimizations, demonstrates zero native bit flips at various temperatures and voltages, proving that high reliability is possible without the aid of additional error correction mechanisms. In a conventional **RO-PUF,** pairs of nominally identical ring oscillators are compared to generate response bits. However, such designs suffer from an inherent limitation: the number of unique CRPs is directly tied to the number of available oscillator pairs, which is bounded by the silicon area. This restriction makes traditional RO-PUFs susceptible to modelling attacks, especially those using machine learning, because the reduced CRP space can be exhaustively profiled.

In contrast to traditional CRO-PUFs, which employ multiplexers to select multiple parallel inverter paths, our architecture directly integrates configurability into the delay element. This method enhances the variability within each stage rather than between different paths. Conventional CROs/RROs alternate between a limited number of pre-established delay paths, resulting in increased routing complexity and additional hardware overhead. However, our approach leverages the dual functionality of XOR/XNOR gates to reconfigure the delay without replicating the entire stages. Instead of selecting among multiple paths, we adjust the delay at the gate level, thereby shifting the configurability from path selection to the behavior of the logic gate itself, which increases the variability within each stage. While classical designs predominantly depend on delay differences between nominally identical oscillator paths, our method introduces configuration-dependent delay dispersion within the XOR/XNOR stages, thereby expanding the challenge space without necessitating additional oscillators. This

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approach also enhances hardware efficiency by eliminating multiplexers and redundant inverter branches, allowing for more configurations per CLB and yielding more CRPS per unit silicon area. From a security perspective, embedding configurability within logic gates enhances resistance to modeling attacks, as the mapping between configuration bits and oscillation frequency becomes more non-linear and challenging to decipher. Finally, because the architecture is defined at the primitive gate level, it is inherently adaptable across technologies, enabling its application in both FPGA and ASIC flows, making it particularly suitable for resource-constrained security applications.

To overcome these limitation, the Configurable and Reconfigurable RO-PUFs architecture was proposed [5]. These approaches introduce additional delay paths, multiplexer-based selection, or reconfiguration logic to expand the CRP space and enhance resistance to modelling. However, they often require significantly more hardware resources, which may not be suitable for lightweight or cost-sensitive applications, such as Internet of Things (IoT) edge devices.
In this study, we propose a configurable delay unit-based modified reconfigurable ring oscillator PUF (MRROPUF) that addresses the CRP scalability problem with minimal hardware overhead and improved modelling resistance and reliability. The proposed design integrates configurable delay elements implemented using XOR and XNOR gates, the configuration of which is dynamically controlled by challenge bits. This approach increases entropy and expands the effective CRP set without proportionally increasing the number of physical oscillators. Security is further enhanced through obfuscated challenge permutations and temporal majority voting, which improve both modelling resistance and reliability.

The proposed MRROPUF was implemented and experimentally validated on Xilinx Virtex-6 and Zynq-7000 FPGAs. The results indicate improvements in uniqueness, uniformity, and reliability compared with existing configurable and reconfigurable RO-PUF designs, while maintaining a low hardware cost. These enhancements demonstrate that the proposed approach provides a practical and efficient means of strengthening hardware security in resource-constrained environments.The results show uniqueness rates of 49.76% and 49%, uniformity rates of 49.3% and 49%, and reliability rates of 99.729c and 99%, respectively outperforming existing configurable and reconfigurable RO-PUF designs while maintaining low hardware cost. These findings demonstrate that the proposed approach offers a practical and efficient route for enhancing hardware security in resource-constrained environments.

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2. Related Work

Three frequently used forms of FPGA-based PUFs are as follows:The arbiter-based PUF or APUF compares the timing or delay information of two identical pathways to select the output [12]. One of the oldest and most thoroughly researched PUF designs is the A-PUF. However, because of the arbiter circuit's (D-Flip Flop) metastability and the delay routes' intricate routing requirements, it has a subpar FPGA implementation [13].The starting value of static random access memory cells (SRAM-PUF) was initially used to introduce memory-based PUF [14]. The primary drawback of these PUFs is that not all FPGA boards have SRAM modules that have not been initialized [14]. All types of FPGA boards may be used to build Butterfly PUF, an alternative memory-based PUF structure [8].In comparison to the A-PUF and Butterfly PUF, the RO- PUF is easy to implement on FPGA. It makes use of the difference in frequency between a pair of identically constructed ring oscillators (ROs). The primary drawback of this type of PUF is that a large number of RO pairs must be included in the design of the RO PUF to provide numerous reliable output bits, which consumes a large number of hardware resources [15]. Due to its need for substantial hardware resources for design, implementing such an RO PUF design poses challenges in many lightweight IoT and embedded system applications. In earlier work, many RO PUF configurations have been presented [16] to reduce the amount of hardware required. In [25], a prediction test screening method that uses fewer hardware resources is suggested as a way to increase the reliability of the ring oscillator PUF. The configurable ring oscillator(CRO) PUF was used in the proposed

authentication protocol and key renewal strategy for clustered sensor networks to increase security [27]. Kursawe et a1. pioneered the development of the Reconfigurable PUF (RPUF) circuit [16], which enhances the number of Challenge- Response Pairs (CRPs) by incorporating various operations through circuit reconfiguration. The reconfiguration signals can modify the PUF behavior either partially or entirely, resulting in the creation of a new PUF. RPUFs can be categorized into physical and logical RPUFs [26]. A notable example of a physical RPUF is the reconfigurable optical PUF, which utilizes a powerful laser to rapidly melt the optical material. This process causes the optical scatter to rearrange, thereby generating new random CRPs. Such a transformation alters the PUF at the molecular level and is irreversible. In contrast, logical RPUFs expand the number of CRPs without changing the original design of the PUF, thereby allowing for reversible modifications.

To reduce resource utilization, an FPGA-based configurable EXOR gate-based reconfigurable RO-PUF was designed. Additionally, the permutation and temporal majority voting (TMV) technique was used to generate the CRPs, which increased the performance of the design. Comparing the proposed PUF architecture to configurable and reconfigurable RO PUF to produce the single-bit response, it is intended to increase CRPs with the same amount of CLB.

2.1. Traditional RO-PUF

The first RO-PUF [9] was created for cost-effective authentication and secret-key generation in cryptography. It includes two N-to-l multiplexers connected to N ring oscillators. In Fig. 1, the multiplexers' outputs connect to two counters, whose values are compared using a comparator. Each Ring oscillator is built with one NAND gate and several inverter gates. Multiplexers select specific ring oscillators, initiate oscillation, and track counter values. After a set time, the comparator checks the counters; if the upper counter is greater, it outputs one.

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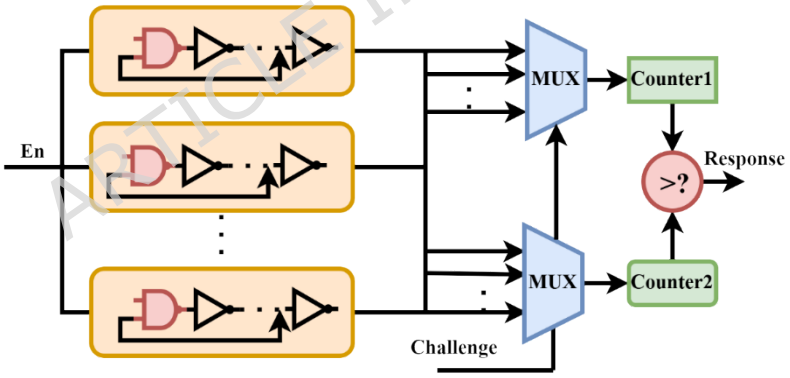


Figure 1: Traditional RO PUF[22].

2.2. Configurable RO(CRO) AND Reconfigurable RO(RRO)

In CRO-PUF, the core logic stage of the RO, typically a single inverter gate, is upgraded to a flexible logic stage capable of handling multiple delay variations. This modification enhances the reliability of the PUF while reducing the hardware requirements [16]. The initial CRO-PUF, introduced in [17], featured eight different configurations achieved by inserting 2:1 multiplexers between each inverter of the RO and providing a single select line for each stage, as shown in Fig. 2. Inspired by this technology, several other CRO-PUF designs, such as those depicted in Fig. 3, have been developed by various researchers [12, 13].

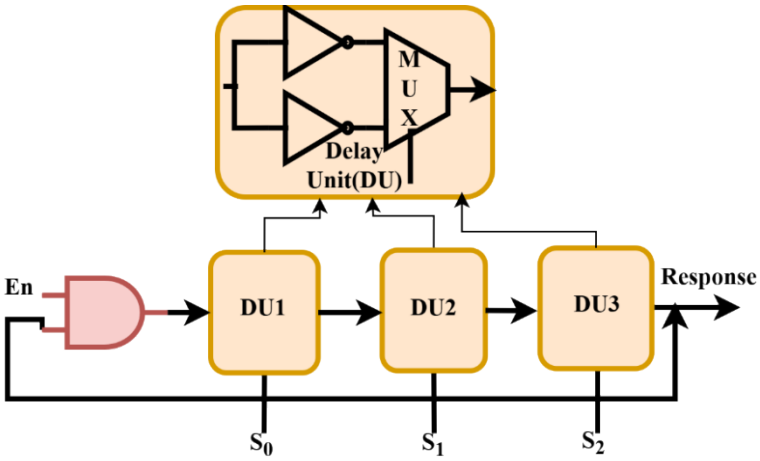


Figure 2: CRO PU F[2].

The CROPUF described in [12] further increases the number of configurations to 256, while maintaining a similar implementation area but with a higher implementation overhead. The selection signals determine which INV contributes to the RO. In the RROPUF architecture presented in [5], as shown in Fig. 4, the programmable nodes N, Nt, ..., S_n of the multiplexers are employed to specify the conduction path for generating an RO. Discrepancies arise owing to changes in the manufacturing of the wires directly connected to the multiplexers.

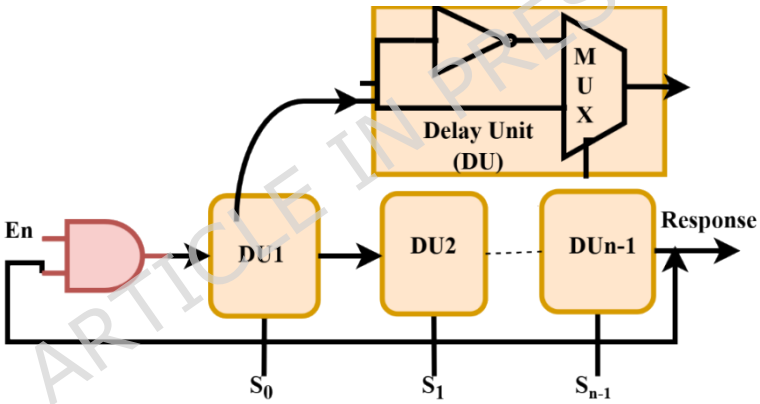


Figure 3: Modified version of CRO PUF[2].

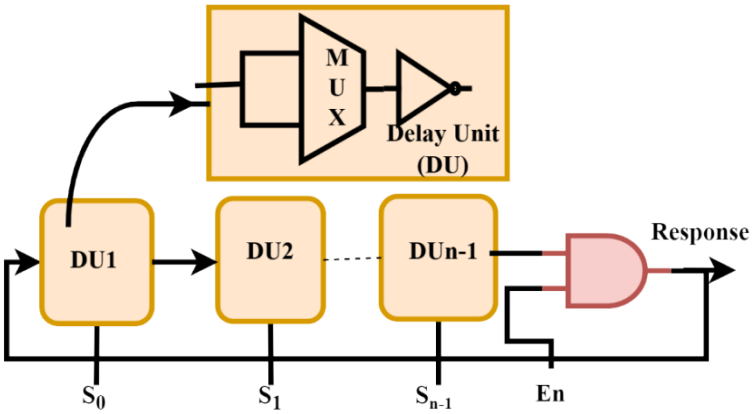


Figure 4: RRO PUF Architecture [5].

In the CRO PUF, the RO unit is designed by introducing MUX units between the NOT gates, thereby altering the RO structure. Although this approach consumes significant hardware resources, it can generate

a larger number of CRPs than a NOT-based RO PUF. Consequently, the CRO PUF [2] and RRO PUF [5] exhibit greater hardware efficiency than a conventional RO PUF [9]. Traditional encryption consumes many resources in IoT devices. Fast and secure authentication is possible because of hardware-based security. However, when the operating environment differs from the standard setup, using an RO PUF may present reliability challenges and security vulnerabilities because of the absence of a sufficient anti-cloning mechanism. To tackle these concerns and reduce hardware expenses, the study introduces an FPGA-based adjustable EOR gate-based Reconfigurable RO PUF, which is designed and implemented. The primary contribution of this research is outlined below. A novel MRRO PUF is designed and implemented to improve the unit resource utilization cost.

Increasing reliability with the use of the TMV scheme and enhancing uniformity with hard macro design methods. A thorough evaluation and comparison with previous ROPUF designs is presented, and the MRRO PUF is implemented on an FPGA. The results demonstrate that, as compared with previous ROPUFs, the implemented designs significantly enhance efficiency in terms of hardware. When compared with configurable and reconfigurable RO PUFs, the proposed PUF architecture is designed to increase CRPs with the same amount of CLB. And also the necessity of PUF in FPGA IP protection is addressed.

This research proposes a unique design method for a low-cost Modified XOR-based Reconfigurable RO PUF (MRROPUF) architecture that takes advantage of the features of EXOR gate logic that works as a converted NOT gate dependent on the appropriate control input. This work also presents the suggested design's implementation on Virtex 6 FPGAs and ZYNQ 7000. The analysis of performance metrics of the proposed work and comparison is carried out with the state of the art.

The subsequent sections of this paper are organized as follows. In Section 3, we provide a detailed description of the configurable EXOR-based MRRO PUF design. In Section 4, we present the characteristics of our proposal, accompanied by comprehensive experimental results. Finally, we conclude the discourse in Section 5 and discussion in Section 6.

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3. Design of Modified Reconfigurable Ring Oscillator PUF(MRRO PUF)

The proposed MRROPUF extensively utilizes Exclusive OR (EXOR) gates. Let the EXOR gate output be denoted as F, with inputs A and B as illustrated in Fig. 5. When A = 0, the EXOR gate operates as a buffer, yielding F = B. Conversely, when A= 1 , the output is inverted,F = B, and the EXOR functions as an inverter. Therefore, the logic function can be switched between a buffer and a NOT gate.



Figure 5: EXOR gate configured into Bulfer or NOT gate based on input A.

The EXOR gate is crucial in the design of the PUF, as it configures the output based on one of the challenge inputs, thereby reducing the prediction rate. A single delay element is constructed using a 2:1 MUX and an EXOR gate, as illustrated in Fig. 6.

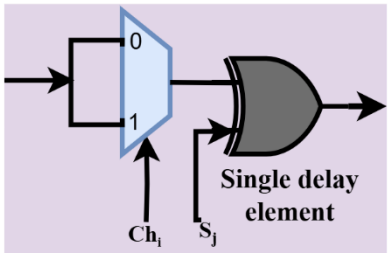
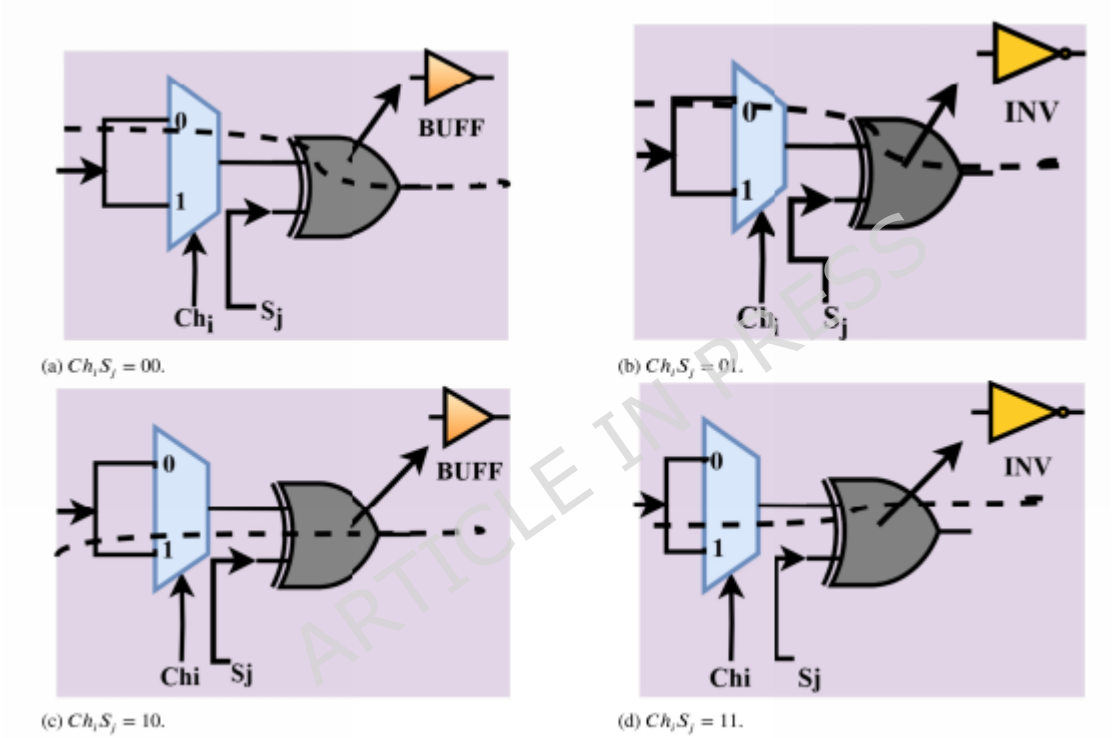


Figure 6: Single delay element of MRRO PUF.

The single delay element consists of two input, one of the input represented as *Ch ¡* is given to the select line of MUX and other *Sj* is one of the two inputs of EXOR gate as shown in Fig. 6. The delay path is dependent on the combination of the C/i; select line of MUX and fij configurable input of EXOR gate. Fig. 7 shows the path with different combinations of *Ch,* and *S* .



(a) $Ch_iS_j = 00$. (b) $Ch_iS_j = 01$. (c) $Ch_iS_j = 10$. (d) $Ch_iS_j = 11$.

Figure 7: EXOR configuration with Delay Unit for Ch_i, and S_J, combinations in single delay element

The two inputs, *Ch;* and *Sj,* to the single delay element yield four distinct combinations. Based on the input *Ch;,* theMUX may select either the 0 or 1 path, and depending on *Sj,* the EXOR gate is configured as either a NOT or a Buffer. This configuration results in four diiferent paths, thereby increasing the number of paths and reducing the prediction rate. The single delay elements are connected sequentially to form a single delay chain. The output of one single delay element is connected to the input of the subsequent element, with four such delay elements connected in series. The output of the fourth delay element is then connected back to the AND gate, collectively forming the single delay chain, as illustrated in Fig. 8.

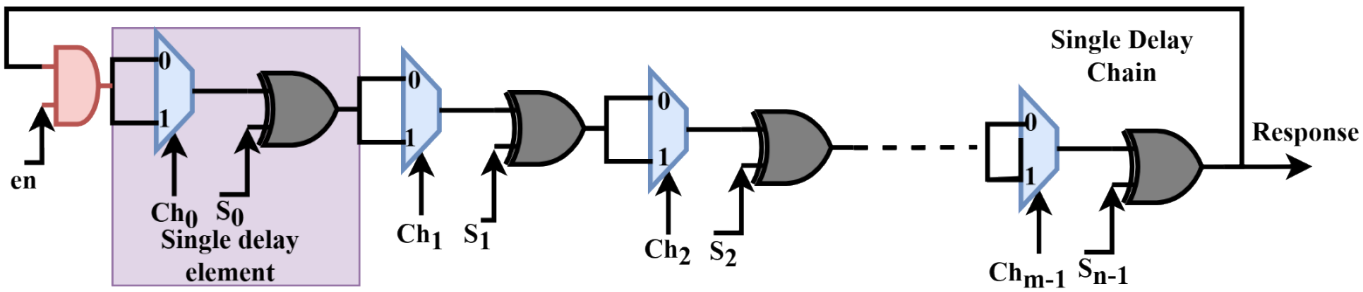


Figure 8: Single delay elements are connected together, forming a single delay chain.

The responses are generated depending on the challenge inputs as given in (4). The equation of the proposed PUF is given by substituting challenge *(C)* in (1). From (1) PUF function can be written as

$$P(C_l) = R \qquad (4)$$

$$\text{Where } C_l = (Ch_i, S_j)$$

$$Ch_i = Ch_0, Ch_1, \ldots, Ch_{m-1}$$

$$S_j = S_0, S_1, \ldots, S_{n-1}$$

$$P\left(\sum_{i=0}^{m-1} Ch_i, \sum_{j=0}^{n-1} S_j\right) = R \qquad (5)$$

The EXOR gate is transformed into a NOT gate by designating one of its two inputs as the configuration input (fi,) in the proposed design, as illustrated in Fig. 9. Consequently, the input EXOR sets $fi_0, S, \ldots, S_n$ are incorporated into the single- bit PUF challenge. Various configuration inputs are applied to the EXOR gates to facilitate their selection and utilization throughout the RO setup. An odd number of EXOR gates is required to form a ring oscillator when these gates are configured as inverters. This configuration signal elicits different responses from various ROs. In this system, the configuration signal is derived from the n-bit challenge, represented as $N_0, S, \ldots, S_n$.

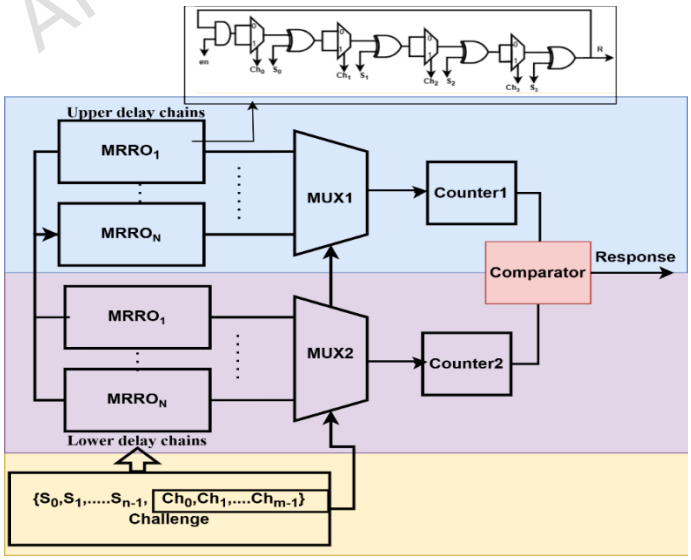


Figure 9: MRRO PUFs present in the lower and upper delay units with input challenges(C) and output response(R).

The final m bits of the challenge are sent to the MUX's choose signal (where 1 = n + m). The MRRO PUF can generate several distinct ROs using this setting strategy. The challenge is used to create a new RO in addition to selecting which ROs to compare; therefore, this configuration option can provide more outputs without utilizing more hardware resources. To increase the number of response bits without using more FPGA resources, the configuration signal in the stimulus can be changed. The AND gate determines when the oscillation begins by utilizing a control signal Enable(En), while the circuit consists o of n identical EXOR gates, each functioning as a programmable inverter.

The signal set fi_0, S_2, ...A,_j is also included in the 1-bit PUF challenge because one of the Exclusive OR (EXOR) gate inputs is used as a delay setup signal, as shown in Fig.8. By applying different configuration signals, the Exclusive OR (EXOR) gates are selected and participate in forming the ring oscillator. For example, in the case of a 4 selection scheme, eight different ROs are possible by setting the configuration nodes as 0,1,1,1, 1,0,1,1, 1,1,0,1, **1,1,1,0,** 1,0,0,0, 0,1,0,0, 0,0,1,0 and 0,0,0,1. The different ROs then provide a range of responses to an identical configuration signal. The n-bit is chosen from the MSB of the challenge bits, and is used in this structure to extract the configuration signal. The final m-bits of the challenge are delivered to the multiplexer's choose signal, where 1 = n + m. (5). The MRRO PUF can be used to construct a wide variety of ROs because of the flexibility of the setup approach. The oscillation frequency of the RO is influenced by several factors, including electromagnetic fields, noise, and ambient temperature. Therefore, the oscillation frequencies will differ even if two ROs are used, even though their properties are identical.

The FPGA tests showed that process differences cause variations in oscillation frequency for the Ring Oscillators (ROs) in different setups. This makes the architecture suitable for creating a PUF. When you change the number of EXORs in the MRROPUF design and use MUX selection bits to select between ’0’ and ’1’ paths, it affects the oscillation frequencies of the ROs. Using more MUX and EXOR stages in the circuit allows building more ROs. Consequently, the inverter (INV) are replaced by an MUX and configurable EXOR gate in traditional RO PUF, creating a MRRO PUF. Each MRRO consists of an AND gate, four MUX, and four EXORs, which are connected together to design the single delay unit, as shown in Fig.8. The control signals represented as S0S1S2S3+ C0C1C2C3 used to configuration EXOR gates and MUX path selection, influencing the start time of the oscillation loop. The "En" input functions as an RO switch. When ”En = 1," the AND gate acts as a buffer, and the circuit's behavior depends on EXOR configuration. With "En = 0," the AND gate output remains low, and the state of the circuit is controlled by the AND gate. This approach, which uses an AND gate, results in a low-power design.

In an RO PUF, many ROs can consume a large amount of power if they continue to oscillate. However, the proposed design does not ensure that all ROs generate responses. By controlling the “En” input, only selected ROs oscillate, while unselected ones stay still, significantly reducing overall power use.

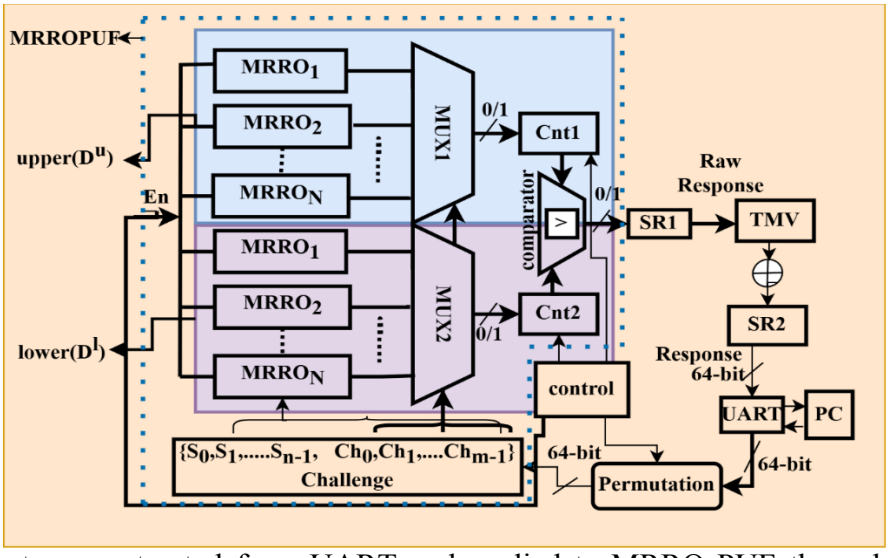


Figure 10: Inputs are extracted from UART and applied to MRRO PUF through permutation and responses are extracted from PC through UART.

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The proposed MRROPUF architecture is based on an XOR-configurable Ring Oscillator(RO) that increases entropy at the cost of minimal hardware overhead. In traditional RO PUFs, each oscillator is based on a fixed delay path, but the proposed design introduces structural configurability at the gate level. XOR gates are used as configurable delay elements, and each RO can operate in multiple structural configurations without additional multiplexers, thus avoiding the additional routing asymmetry and deterministic delay components introduced by multiplexers. In contrast to CRO and RRO PUFs, where configurability is implemented by external path selection, such flexibility is moved inside the LUTs, and by perturbing delays at the LUT leve the design can finely tune the internal propagation paths, making the circuit more sensitive to manufacturing variations and increasing the structural diversity within a single CLB.

The pseudo-algorithm is presented to generate the 64-bits response from the proposed architecture. Fig. 10 shows the main architecture to generate the responses from proposed MRRO PUF design.Two RO are engaged in producing the response from the configuration signals N S N_2N ...N,_t and C_OC $C_2C_3...C_{m_1}$to the MRROs are determined by the excitation signal.

Each RO's output signal is routed to the MUX output, which is then connected to the rising edge counter's trigger port. The counter value rises by one when the oscillation signal's rising edge appears.The ring oscillator begins to oscillate once the control module enables the RO pair. The enable signal is set to "0" when the count reaches a certain number, and the RO pairs cease oscillating as a result. The comparator is now supplied the count value from the two counters, "Cntl " and "Cnt2." The response is "1" if "Cntl > Cnt2," which means that MRRO1 occurs more frequently than MRRO2; otherwise, the response is "0." To obtain the 64-bit output response, the 1-bit answer is an input to the shift register 1 (SRI).

Algorithm 1 MRRO PUF Algorithm

```
MRRO_PUF -+ FPGA(8 rows, 8 columns,
64 CLBs) (RO present in the CLBs)
Master Challenge +- UART
Receive(64 bits) (Input from the
UART}
Permutated_Challenge +-
Permutation_Function (Obfuscated input
to the ring oscillator array}
Sub_Chal1enges +- Divide Permutated_Chal1enge
into 8-bits (Permutated challenge divided into eight 8-bit
values total of 64-bit} for each Sub_Challenge do
  RO_Group1=MUX1[(Sub Challenge LSBs MUX RO) + (Sub Challenge
  MSBs Ex RO)] RO Group2=MUX2[(Sub Challenge LSBs MUX RO)+(Sub
  Challenge MSBs EXOR RO)] if En == 1 then
    Frequency_Group1=Measure_Frequency(RO_Group
    1)
    Frequency_Group2=Measure_Frequency(RO_Group2
    )
  e
nd
if
end
for
if 9-bits_ref_cnt() starts then
  Counter I (Cntl) and Counter2(Cnt2)
start end if
if 9-bits ref cnt == max value then
  En = 0
  Cntl and Cnt2 stop
```

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```
counting end if
if Cntl > Cnt2 then
  Response(R) = 1
else
  Response(R)
= 0 end if
64 bits ShiftRegister1 = Raw Responses | Store raw responses in a 64-bit
shift register if Enhanced Response Bits=Temporal Majority Voting
(Response Bits, 64) then
  golden response=1
else
  golden_respon
se=0 end if
Golden Responses=Determine Golden
Responses (Raw Responses Register)
Final Response=Bitwise XOR
(64_bits_Golden_Responses)
Final Response 64 bits ShiftRegister2=Store(Final
Response)
```

```
Send_Response_To_User_PC(UART_Connection,
Fina1_Response) Control_ROs(Enable_Input)
```

According to Fig. 10, the RO-PUF is initially implemented by initiating a 64-bit master challenge from a user PC (Matlab) over a UART interface, which is then applied to a permutation function to produce a permutated challenge. The data encryption standard (DES) cryptographic technique is used by the permutation function to reorder the words in a series. The MRROPUF implementation uses only the fixed, public DES Initial Permutation (IP) table a keyless, combinational 64 bit remapping, requiring just FPGA LUT6 primitives, not full DES encryption with rounds, S-boxes, or key management. The 64-bit input challenge undergoes static rewiring (e.g., bit 58 position 1, bit 50 position 2) before feeding three PUFs, rotating the feature space from f(C_in) to f(DES_IP(C_in)) to disrupt linear ML modeling. No key exists the permutation is hardcoded at synthesis across all 10 FPGAs, with zero key-related signals in the Vivado netlist. Our adversary model assumes full attacker knowledge: they know/invert the public DES IP table and attack using actual permuted challenges collected from the 100000.

The internal challenge bits are basically concealed by this permutation strategy. Then, two distinct ROs are selected for comparison from each of these sub-challenges (permutation output). The least significant bits (LSBs) in the challenges choose one of the eight ROs in group 1 and group 2 using multiplexers 1 (MUXI) and 2. (MUX2). The PUF is provided the whole challenge sequence in order to set the EOR to the specified number of NOT gates and select the path through the MUX. The frequency of the chosen ROs from groups 1 and 2 is then acquired and supplied into the corresponding 9-bit counters.

The 9-bit reference counter is powered by a crystal clock signal produced by an on-board oscillator at 100 MHz (controller). The reference counter, counter 1, and counter 2 begin counting simultaneously and are compelled to stop when the reference counter reaches its maximum value. The response bit for this RO pair is then generated by comparing the values of counters 1 and 2, depending on which counter has the greater value. In this study, Temporal Majority Voting (TMV) was used before the EOR operation to enhance attack complexity and decrease response instability. The inputs for delay control are applied 64 times to each level. The raw responses are stored in a shift register with 64 bits.

The "golden responses" is then determined for them using the TMV principle. If more than 50% of the raw replies generated are 1s, the golden response is deemed 1, else it is considered 0. A 1-bit "final response" is produced by EXORing the 64 1-bit generated golden replies (TMV output), which are stored in a 64-bit shift register. Each 64-bit master challenge (challenge from UART) generates a total of 64-bit final

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response bits, which are then placed in a 64-bit shift register. The final response bits are sent to the PC through a UART 8-bit connection in order to be analysed for **PUF** performance. The ROs are chosen, the oscillations are counted, and responses are returned based on comparisons by the controller circuitry. It utilizes the enable input to start and stop the ROs.

To understand the behavior of the delay PUF, it is essential to understand the concept of the delay model. Each route has two delay components: a fixed delay component and a variable delay component. The overall delay in a circuit is estimated by summing the delays imposed by individual gates and connections for all routes. Owing to changes in production techniques, a variable delay component was added. In an ideal world, differences in the production process should have the greatest effect on the PUF output. As a result, the variable delay element is more important in a PUF than the fixed delay element. Because the RO-PUF is built on delay-based concepts, we've defined a net delay, abbreviated as D given in (6).

$$Delay(D) = D_S + D_R .\#(6)$$

D_S and D_R represent static and random delays.

For proposed design The delay equation is given by (2)

$$D_{MRRO} = (D_S + D_R)_{MUX} + (D_S + D_R)_{EXOR} .\#(7)$$

$(D_S + D_R)_{MUX}$ delay due to MUX.
$(D_S + D_R)_{EXOR}$ delay due to EXOR.

A small contribution from the static delay component can lead to an unbalanced PUF output. When $(D_S)_{MUX} + (D_S)_{EXOR} > (D_R)_{MUX} + (D_R)_{EXOR}$, random variation does not affect the output, and the static delay dictates the PUF output. This results in a skewed PUF output.

For instance, Ch_i is i^{th} delay unit of challenge input that is selection signal $Ch_i(i = 0,1,2,3....m -1)$ is "0" is different from the value when Chi is "1," as illustrated in Fig.7 due to manufacturing variances, the wires and the multiplexer directly attached to the MUX. Effective Delays (EDs) are the delays for paths "0" and "1," which are represented by the letters ED0 and ED1, respectively. The total of the transmission gate and wire delays is known as an ED. When the control input is 0, the EXOR gate functions as a buffer, and when it is 1, it functions as an inviter. Every delay unit is equally routed, but EDs between pairs may change due to variations. Paths "0" and "1" are identically routed. A MRRO is an excellent choice for building a PUF because of its unique configurable property.
The reliability is determined by the delay difference between chain of delay unit, to generate a single bit response. Reliable bits can be obtained by maintaining the large delay difference between two delay chains. The total delay equation of MRROPUF is given in (8)

$$D_{MRRO} = (D_S + D_R)_{MUXpath0/1} + (D_S + D_R)_{EXOR} .\#(8)$$

From the Fig.8 the delay difference between the MUX and the EXOR by delay path0 and delay path1 is given in (9)

$$D_{diff} = D_{MUX0} - D_{MUX1} .\#(9)$$

D_{M0} and D_{M1} delay of path0 and path1 of MUX respectively.

To generate the 1-bit PUF response, single delay units are connected in series to form a delay chain, the figure Shown in Fig. 8such chains are placed parallelly as upper(DRO^u) and lower(DRO^l)groups then the comparing the delay difference between the delay chains from upper group and lower group that is represented in (10)(11).

$$DRO^u = D^u_{diff} = D^u_{MUX0} - D^u_{MUX1} + D^u_{EXOR}.\#(10)$$

$$DRO^l = D^l_{diff} = D^l_{MUX0} - D^l_{MUX1} + D^l_{EXOR}.\#(11)$$

Single bit response is generated depend on the delay difference between upper and lower delay chain is given in (12)

$$(DRO^u - DRO^l)\begin{cases} > 0, & R = 1 \\ < 0, & R = 0 \end{cases}\#(12)$$

For multi-bit response, the single delay chain is replicated in the upper and the lower groups as given in (13) and (14). Then (12) is rewritten, and is given in (15).

$$DRO^u_i = DRO^u_1 + DRO^u_2 + DRO^u_3 + \cdots + DRO^u_n\#(13)$$

$$DRO^l_i = DRO^l_1 + DRO^l_2 + DRO^l_3 + \cdots + DRO^l_n\#(14)$$

$$\sum_{i=1}^{n} (DRO^u_i - DRO^l_i)\begin{cases} > 0 & R = 1 \\ < 0 & R = 0 \end{cases}\#(15)$$

DRO^u_i and DRO^l_i are the single delay chains present in the upper and lower group as shown in Fig.10.
When addressing delay-based PUFs, each switch can be efficiently represented using only two parameters. When the switches are linked in series, the neighbouring parameters can be combined and stated by a new parameter. The new parameter is denoted by $\delta$and is known as the path segment delay. As a result, the PUF challenge/response relationship can be formalised as in (16)

$$\sum_{i=1}^{N} (-1)^{\rho_i^N}\delta_i + \delta_{N+1}\begin{cases} > 0 & R = 1 \\ < 0 & R = 0 \end{cases}\#(16)$$

Where ρ^j_i is given as

$$\rho^j_i = \bigoplus_{x=i,i+1,\ldots,j} C_x = C_i \oplus C_{i+1} \oplus \cdots \oplus C_j\#(17)$$

The symbol δ denotes the delay difference of the ρth stage, N is the bit length of the challenge, e.g. $N = 64$ to generate $64 - bit$ output response.The valur of response output is '0', if the total delay is negative, otherwise response output is '1'.
The challenges to $\rho's(i \leq j)$. $\bigoplus$ and $\oplus$ denote the parity generation and \textcolor{red}{\hl{exclusive-or(XOR)}} operations respectively; C_i corresponds to

the i^{th}challenge bit in the challenge vector $\vec{C} \in \theta^N$, $\theta = \{0,1\}$. By acquiring large CRPs from the PUF, an opponent Eve might construct up a model and estimate the value of all CRPs. The above model can be utilised

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to express the intended PUF design's delay characteristics. The only difference between them is that the path segment delay δ of the model of the previous PUF design originates from a NOT gates; however, the path segment delay δ^* of the proposed PUF includes not only the delay of the MUX but also Exclusive OR (EOR). Thus, the proposed PUF challenge and response relationship can be expressed by (18)

$$\sum_{i=1}^{N} (-1)^{\rho_i^N} \delta_i^* + \delta_{N+1}^* \begin{cases} > 0 & R = 1 \\ < 0 & R = 0 \end{cases} \#(18)$$

Where

$$\delta^* = \delta_{y,M} + \delta_{z,EOR} \text{ such that } y,z \leq N$$

As the complexity of the path segment delay increases, so does the complexity of each 1-bit response generation circuit. According to the above circuit analysis, the suggested PUF more complex than the prior PUF design.

Fig.10 depicts the top module of the MRRO PUF. Using the FPGA, eight rows and eight columns of CLBs are employed to create 64 MRROs. According to the location, the MRROs' frequencies are contrasted. Each MRROs placement is set within a single, two-slice CLB. Two FPGAs, the ZYNQ 7000 and Virtex 6, are used for the implementation, and the results are analysed in the section on result analysis.

4.EXPERIMENTAL RESULTS

4.1IMPLEMENTATION

Typically, FPGAs are utilized as platforms for verification to evaluate the effectiveness of PUFs. The proposed MRRO PUF is implemented using ZYNQ 7000 (Artix-7) and Virte-6 FPGAs. There are sufficient CLBs in this FPGA; each configurable logic block (CLB) has 6 MUXs, 16 D-flip-flops, and 8 LUTs. In this article, a MRRO PUF is designed utilising four MUX, four EOR gates, one AND gate, and a 2:1MUX configuration, which calls for eight LUTs in a CLB of the Artix-7 and Virtex- 6 FPGAs and one MUX7 represented in Fig. 11 The FPGA tests showed that process differences cause variations in oscillation frequency for the Ring Oscillators (ROs) in different setups. This makes the architecture suitable for creating a PUF. When you change the number of EXORs in the MRROPUF design and use MUX selection bits to select between ’0’ and ’ 1’ paths, it affects the oscillation frequencies of the ROs. Using more MUX and EXOR stages in the circuit allows building more ROs.

Consequently, the inverter (INV) are replaced by an MUX and configurable EXOR gate in traditional RO PUF, creating a MRRO PUF. Each MRRO consists of an AND gate, four MUX and four EORs, are connected together to designed the single delay chain in Fig. 11.

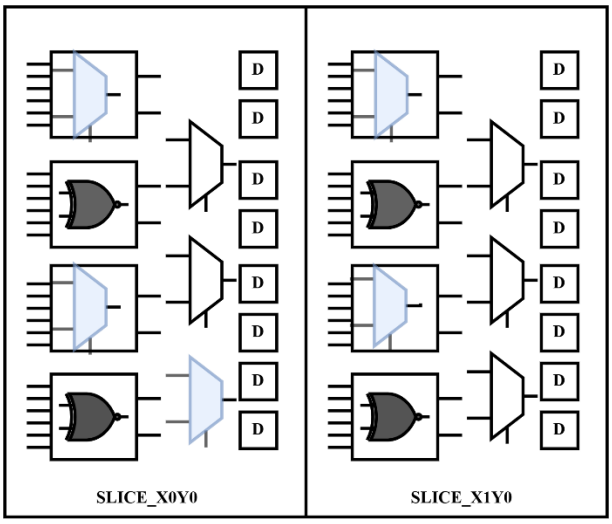


Figure 11: Representing the single delay chain of MRRO in CLB.

The control signals represented as S_0Nt N_2Nj, C_0Ct C_2C_3 used to configure EXOR gates and MUX path selection, influencing the start time of the oscillation loop. The "En" input works as RO switch. When "En = 1," the AND gate acts as a buffer, and the circuit's behavior depends on EXOR
configuration. With "En = 0," the AND gate output remains low, and the circuit's state is controlled by the AND gate. This approach, using the AND gate, results in a low-power design.In an RO PUF, many ROs can consume a lot of power if they all keep oscillating. However, the proposed design doesn't make all ROs generate responses. By controlling the “En” input, only selected ROs oscillate, while unselected ones stay still, significantly reducing overall power use. The proposed design better utilizes the FPGA's LUT resources and it incorporates configurability directly into the XOR- based logic. It does not rely on additional multiplexers, because unlike CRO PUFs, it does not require dedicated MUXs to choose between different delay paths. Rather, the XOR gate serves as both an inverter and a configurable delay, thus there is no additional selection logic. Furthermore, the design exploits the "don't-care" inputs of FPGA LUTs, so it can be optimized for better performance. The inputs can be configured accordingly, and the internal signal routing within the LUTs is slightly altered. This provides very fine-grained delay control at sub-gate granularity, because a large number of oscillator configurations can be realized within the same physical implementation. This expands the CRP space without additional area overhead, making the architecture more efficient. Finally, the architecture enables selection of RO pairs with adequate frequency separation, and if the oscillators have large frequency margins when paired, the design will perform better. The responses will be more robust against variations in temperature and voltage; therefore, they will be more reliable.

4.2 **RESULTS and DISCUSSION**

Ten ZYNQ 7000 and Virtex 6 are used to design and implement the proposed MRRO PUF designs, with an temperature range of 0°C to 70°C. Reliability measures the PUF steadiness at various temperatures, whereas uniqueness measures the capacity to distinguish between two dissimilar devices. Consider the Hamming distance (HD) between the PUFs response to determine both uniqueness and reliability. Hamming Weights calculates uniformity. Performance metrics for the proposed MRRO PUF are examined and contrasted with those of earlier PUFs based on FPGAs.

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4.2.1Uniqueness

It refers to the extent of variation in the responses generated by a PUF when implemented on two different chips, while being subjected to the same single challenge. In order to quantify this variation, the concept of inter-Hamming distance is used. The inter-Hamming distance, also known as uniqueness, is a performance parameter that computes the difference in the responses of PUFs implemented on various chips, when presented with identical challenges.Uniqueness of the proposed PUF is calculated using (19)

$$\text{Uniqueness} = \frac{2}{K(K-1)} \sum_{i=1}^{K-1} \sum_{j=i+1}^{K} \frac{HD(P_i(C_l)'P_j(C_l))}{n} X100\% \qquad (19)$$

k represemnts number of devices, $P,\{C$) is the i'/t device response, and P,(C) is the jth device response to the same challenge C . The greatest variation in the responses of two PUFs should be indicated by a uniqueness value that is close to 50%. It indicates the unpredictability of the responses, hence it also has an impact on a PUF's security. Fig. 12 is an example of assessing uniqueness.

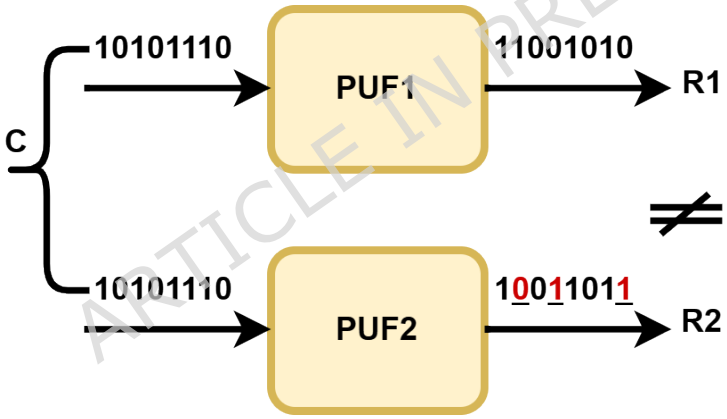


Figure 12: Uniqueness of the PUF with same challenge(C) input given to the two PUF designs on two devices of the same type outputs different responses(R).

The distinctiveness of the proposed PUF is assessed by implementing the design on Virtex 6 and ZYNQ 7000, and by manually placing the PUF design in five different locations on the two FPGAs. The average uniqueness is determined by evaluating responses generated from the design positioned identically on both FPGAs, with the resulting graph depicted in Fig. 13 and Fig. 15, respectively. The average uniqueness of responses from the ten different placements on Virtex 6 and ZYNQ 7000 is illustrated in Fig. 14 and Fig. 16, respectively.

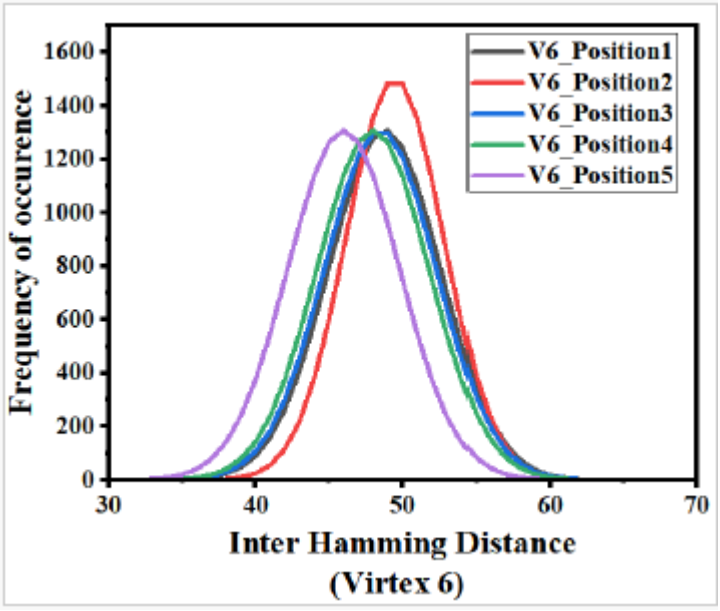


Figure 13: Uniqueness of responses generated, when design place at five diiferent places in FPGA.

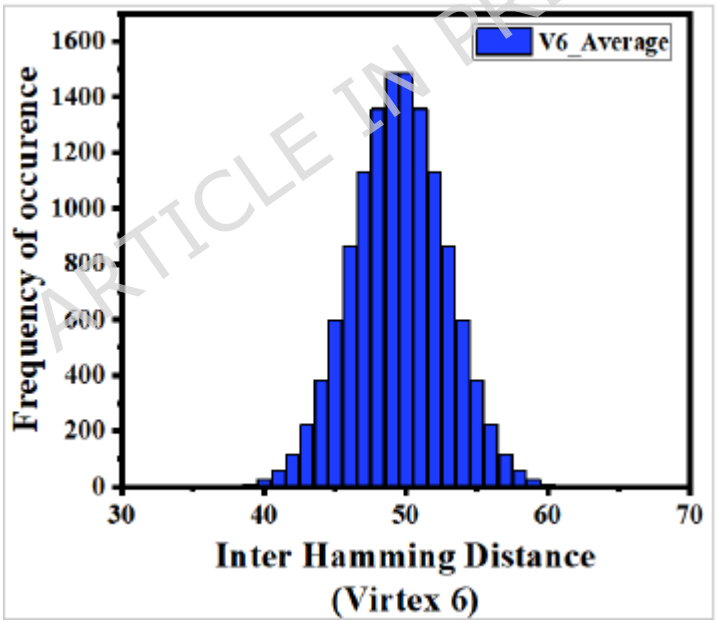


Figure 14: Average uniqueness of MRRO PUF evaluated from responses obtained from 10 difference in FPGA

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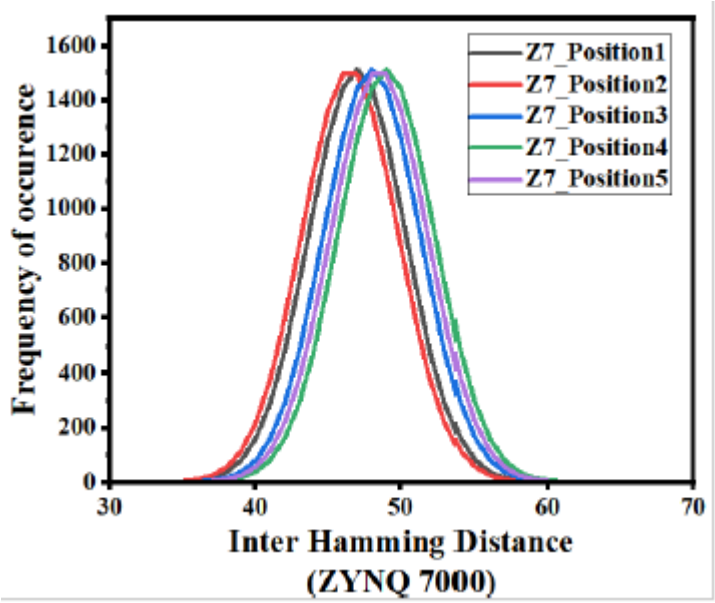


Figure 15: Uniqueness of the PUF design placed at ten different positions in ZYNQ 7000 FPGA.

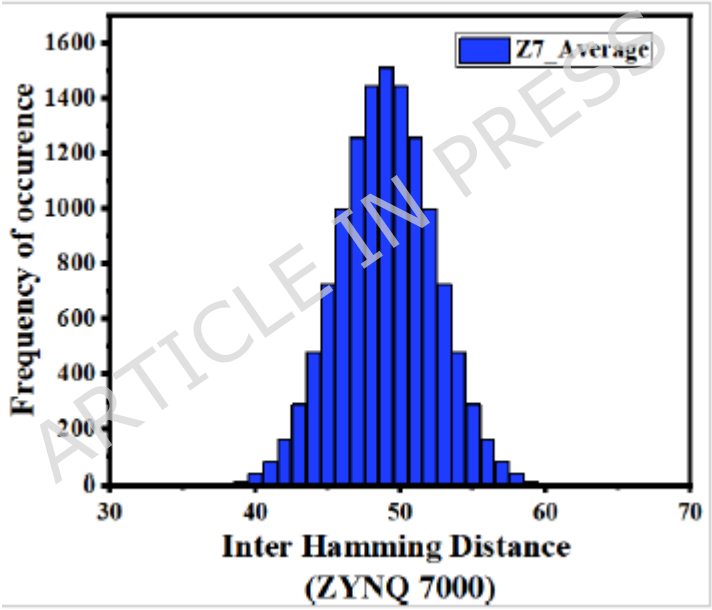


Figure 16: Average uniqueness of calculation from five different positions in ZYNQ 7000.

The proposed MRROPUF is implemented on ten Virtex 6 and ten Zynq 7000 devices. Uniqueness is assessed by applying 1500 CRPs to generate the corresponding 64-bit responses with a 64-bit challenge. The uniqueness values, calculated using equation (19) for the proposed MRROPUF, are 49.76% and 49%, respectively, which are extremely close to the optimal value of 50%.

4.2.2 Reliability

The Hamming distance of a PUF under varied environmental circumstances is a measure of its reliability. Variable temperatures are typically considered when evaluating a PUF's reliability. Different temperature gradients were used to measure the PUF chip responses. A PUF should, ideally, respond to challenges in the same way each time. A PUF output is intrinsically stable to

changes in the effective environment. Since PUFs extract minute variations in chips. Reliability measures the stability of the PUF, and a value of l00% implies that the PUF response value is consistent.and that under all test conditions, the response sequence contains no bit flips. The reliability formula is given by (20)

$$R = \left(1 - \frac{1}{x}\sum_{y=1}^{x}\frac{HD(P(C_i),P(C_{i,y}))}{n}\right)X100\% \qquad (20)$$

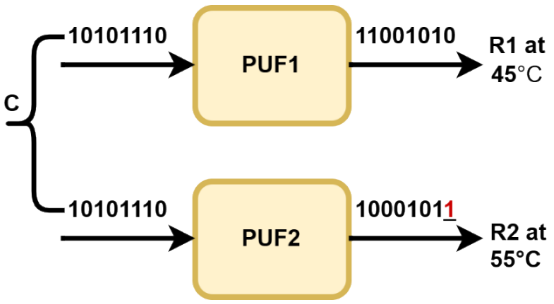


Figure 17: Reliability of the PUF with same challenge(C) input at different temperatures results in two responses(R) with single bit change.

where m is the number of tests and n is the number of positions in the response's sequence. *P(C, t)* ' *t'h* sample of *P(C,)* among recurring responses. Under various environmental conditions, the reliability is calculated using the formula given in (20). The reliability evaluation example with varying temperature and the same challenge is displayed in Fig. 17. The reliability of the proposed PUF is measured by implementing the design in Virtex 6 and varying the temperature for same input challenges.Repeating the process by manual placement of the PUF design in the five different positions of the FPGAs and the resultant graph is shown in Fig. 18, Fig. 19.

The reliability of the proposed PUF is measured by implementing the design in ZYNZ 7000 and varying the temperature for same input challenges.Repeating the process by manual placement of the PUF design in the five different positions of the FPGAs and the resultant graph is shown in Fig. 20, Fig. 21.

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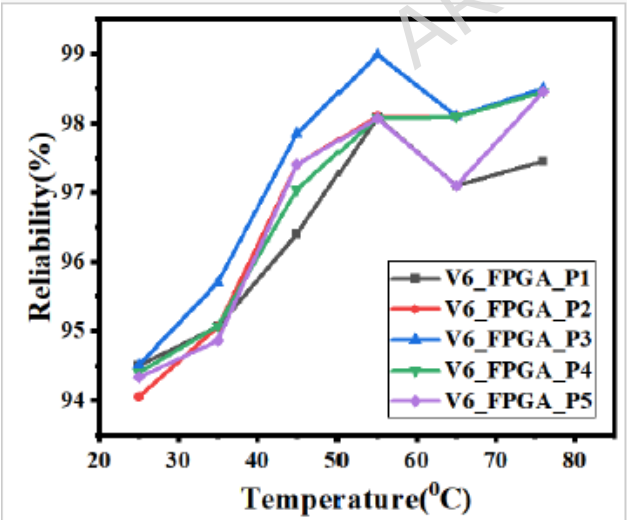


Figure18:Reliability for varying temperature on Virtex 6.

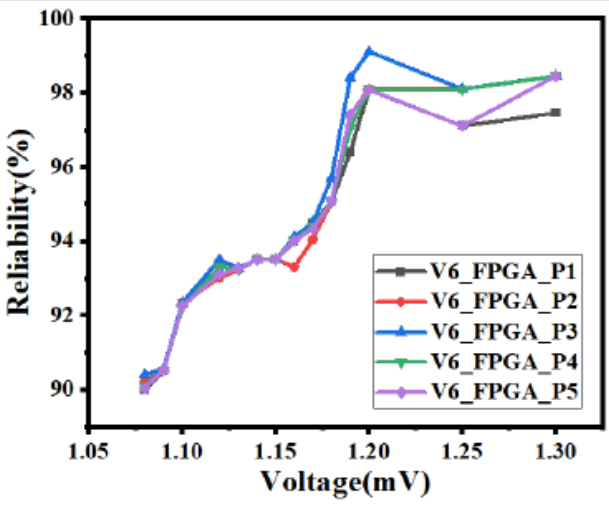


Figure 19: Reliability for varying voltages on Virtex 6.

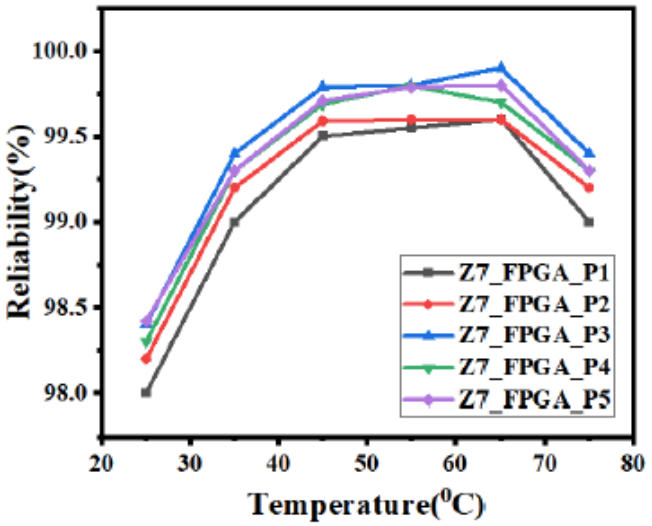


Figure 20: Reliability for varying temperature on ZYNQ 7000 FPGA.

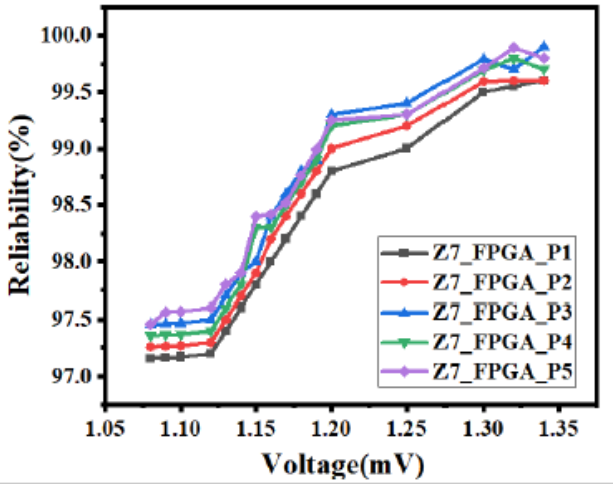


Figure 21: Reliability for varying voltage on ZYNQ 7000.

The reliability of the proposed design is calculated with 1500 CRPs with varying temperature between 0°C to 70°C in step of l0°C, when implemente on ten Vertex6 and ten ZYNQ 7000 FPGA, responses are generated resulted into 99.729a and 99%, respectively, for vary voltage and temperature.

4.2.3 Uniformity

The properties of the response space are revealed by uniformity, where uniformity evaluates the total bias level of a series of 64 bit response. In Fig. 22, gives the stimulus to response pair formed by the MRRO PUF design is given as the ratio of 0 and 1 in each PUF output sequence. The optimal values for uniformity is 50%. The distribution of the 0's and 1's is calculated using (21). In this case, $r_{i,j}$ is the j^{th}bit of the nth chip's n-bit response.In this case, r_i, is the y_i/t bit of the nth chip's nbit response.

$$\text{Uniformit } y_i = \left(\frac{1}{n}\sum_{j=1}^{n} r_{ij}\right)X100\% \qquad (21)$$

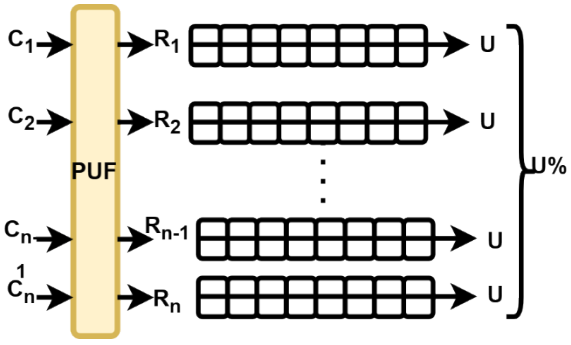


Figure 22: Uniformity for different input challenges with corresponding responses.

To determine uniformity, the suggested MRRO PUF architecture with the specified EXOR gate was implemented on ten ZYNQ 7000 and ten Virtex-6 FPGAs. Equation (21) was used to derive the uniformity value for the MRRO PUF, which was calculated to be 49.39c and 49.9â on Virtex6 and ZYNQ

7000, respectively, which are close to the ideal value of 50%. The percentage distribution of the 0's and 1's of proposed PUF for Vertex6 and ZYNQ 7000 is shown in Fig. 23 and Fig. 24 respectively.

4.2.4Entropy Analysis

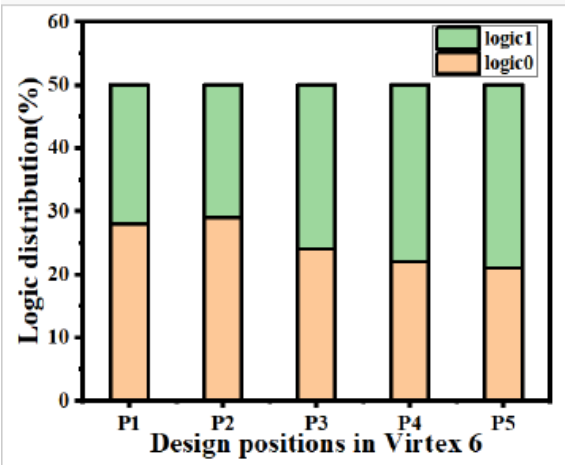


Figure 23: Logic distribution on Virtex 6.

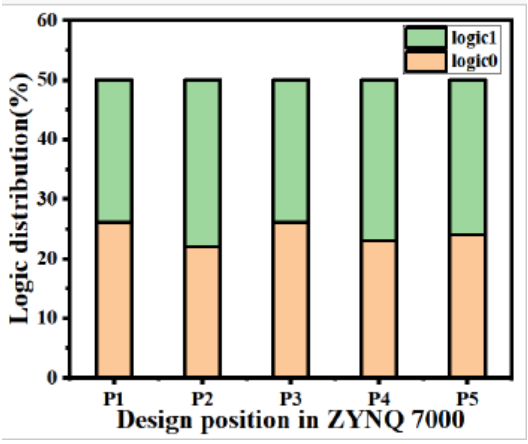


Figure 24: Logic distribution on ZYNQ 7000.

Entropy analysis is frequently used as a response to assess the unpredictable nature of PUF. To measure the unpredictable nature of the random response, the min-entropy evaluates the lower bound of entropy (i.e., the worst-case scenario). p_1 and p_0 are the probabilities of occurrence for the values of "1" and "0," respectively, considered for *k* devices for each bit in the *n*-bit responses.

p_1 and p_0 are calculated by HW_i/k and $1 - HW_i/k$ respectively, where HW_i calculates the number of 1s in *k* devices. When $p_{i\,max}$ is the maximum of these two probabilities ($p_{i\,max} = \max(p_1,p_0)$), the definition of the minimum entropy of each individual bit (binary random sequence) is given in Equation 22, the overall minimum entropy, as explained in :

$$H_{min,i} = -\log_2(p_{i\,max}) \#(22)$$

$$p_{i\,max} = \begin{cases} \frac{HW_i}{k}, & \text{if } \frac{HW_i}{k} > \frac{k}{2} \\ 1 - \frac{HW_i}{k}, & \text{otherwise} \end{cases} \#(23)$$

$$(H_{min})_{average} = \frac{1}{n}\sum_{i=1}^{n} H_{min,i} \#(24)$$

Table 1: Entropy values for different FPGA types

FPGA type	Entropy
Virtex 6	0.772
ZYNQ 7000	0.812

It can also be deduced from Equations 23 and 24 that the total average minimum entropy is given by Equation 24. In this research, the average minimum entropy of 10 PUFs implemented by two sets of FPGAs was calculated using equations 22 and 24. The average entropy of the proposed MRRO PUF in

the two different types of FPGAs is clearly visible in Table 1 to be sufficiently near to 0.792. The suggested MRRO PUF's entropy value is estimated conservatively using the NIST SP800-90B test because it could increase if more FPGAs are added .

4.2.5 NIST Test

Randomness refers to a characteristic where the sequence produced by the designed PUF circuit exhibits high levels of entropy and uniform distribution. Greater randomness correlates with enhanced security, making it more challenging for potential attacks. For testing randomness, the NIST SP80022 test package offered by the National Institute of Standards and Technology is frequently utilized. The values p and p_0 are determined by the formulas $H W,/k$ and $1 — H M,/k,$ respectively, where $HP,$ counts the number of 1s across k devices. When $p;qpg$ represents the greater of the two probabilities $(p;q_a — max\{p , p0)),$ the definition of the minimum entropy for each individual bit (binary random sequence) is described by equation (23). The overall minimum entropy, as detailed in [26], is utilized for statistical testing [32]. To better quantify the randomness, we employed NIST SP800-22 to evaluate the random sequence produced by the development board. The version of the NIST test software utilized in this study is STS 2.1.2. For the MRRO PUF implementation detailed in this research, the output of the 10*64-bit random sequences on the Virtex- 6 and ZYNQ 7000 FPGA development boards are illustrated in Fig. 25 and Fig. 26, respectively. The p-values obtained from the test results exceed the established significance threshold of 0.01. The ten tests indicate that the PUF introduced in this research demonstrates relatively good randomness, is quite resilient against attacks, and is suitable for use as a chip encryption technology.

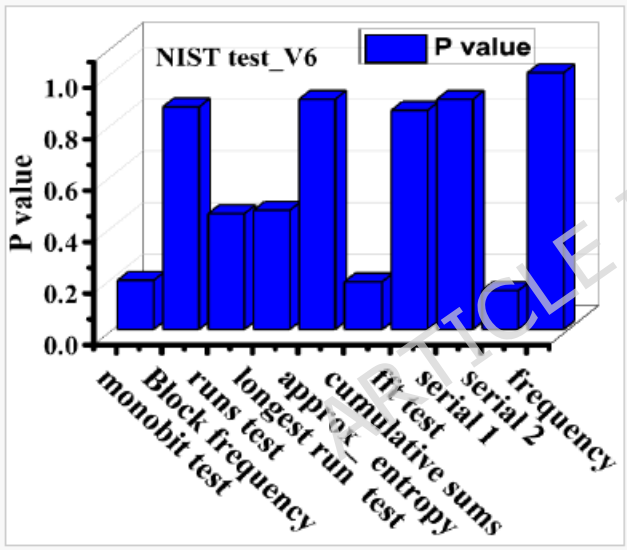


Figure 25: NIST test for responses generated from Virtex 6.

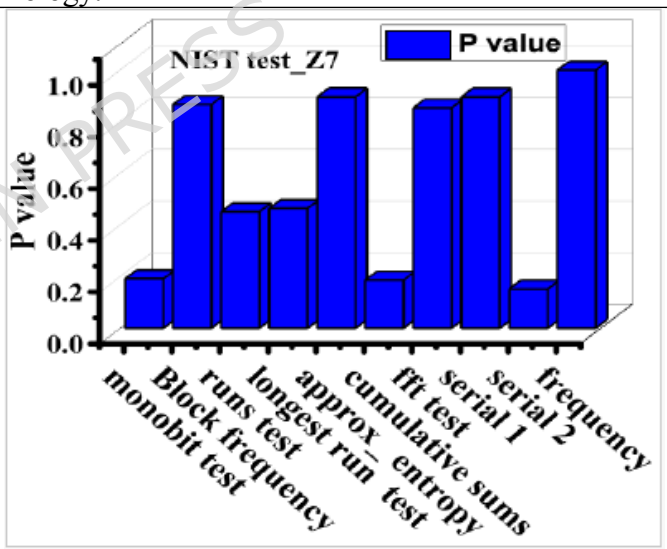


Figure 26: NIST test for responses generated from ZYNQ 7000.

The proposed design evaluation involved analyzing 1500 x 64-bit MRROPUF response sequences per Field-Programmable Gate Array (FPGA), culminating in a total of 9600 bits per device across 10 Virtex-6 and Zynq-7000 devices. The bitstream construction process entailed generating responses from unique 64-bit challenges under controlled reference conditions, specifically maintaining a supply voltage of 1.0 V and an ambient temperature of 25°C, to minimize environmental artifacts. For concatenation, the standard PUF evaluation protocol was followed, whereby all responses were sequentially concatenated across devices to form a single test sequence of approximately 96000 bits (10 FPGAs x 96,000 bits). This methodology aligns with established PUF characterization standards as documented in prior literature, such as the proposed MRROPUF, validates SP 800-22 on concatenated 1500 x 64-bit sequences with documented passing results.

Bitstreams for NIST SP 800-22 tests are formed by concatenating responses from 100,000 unique CRPs per device (2^{64} challenge space), yielding 100,000-bit streams. This meets NIST minimums (e.g., 100 trials of 100 bits for most tests), avoiding power loss seen in shorter streams per our simulations. P-values pass if 0.01<p<0.990.01<p<0.99 (one-sided) or via trial proportions (multi-sequence), with averages, confidence intervals from 10 samplings/device.

4.2.6 Modelling attack

Among the many threats to Physically unclonable functions, machine learning modelling attacks have emerged as a popular and well-studied type of threat. The evaluation of the proposed PUF in terms of ML attacks is being employed with several ML attacked methods like Decision Tree(DT), Support Vector Machines(SVM), Covariance Matrix Adaptation Evolution Strategy(CMA-ES) Random Forest(RF), K-nearest Neighbor(KNN). Several experiments were conducted with varying training sample sizes in this study. Thus, we evaluated the vulnerability of the proposed MRROPUF design to modelling attacks based on machine learning algorithms.

The goal of the vulnerability analysis is to assess the robustness of the diiferent RO-PUF designs against machine learning- based modeling attacks, and the results of this study are expected to provide insights into designing more robust PUFs in the future. For the assumed threat model, the attacker has access to a subset of the MRRO-PUF CRPs, namely 25% of the CRPs, and for calculating the predictive accuracy of the machine learning models, the standard split is used, i.e., 75% of the data is used for training and 25% of the data is kept aside for testing. Each MRRO-PUF CRP covers 75% of the total CRP space, and 25% of the MRRO-PUF CRPs are included in each randomly chosen set of 100,000 CRPs used to train the different machine learning models, which allows five of the most commonly used machine learning algorithms to be used to evaluate the susceptibility of the RO-PUF design to modeling attacks in a comprehensive manner and thus allows for a broad comparison of how well an adversary can learn and predict the behavior of the PUF.

The results of the ML modelling attack evaluation on the CRPS generated from the Virtex 6 and Zynq 7000 FPGAs are shown in Fig. 27.

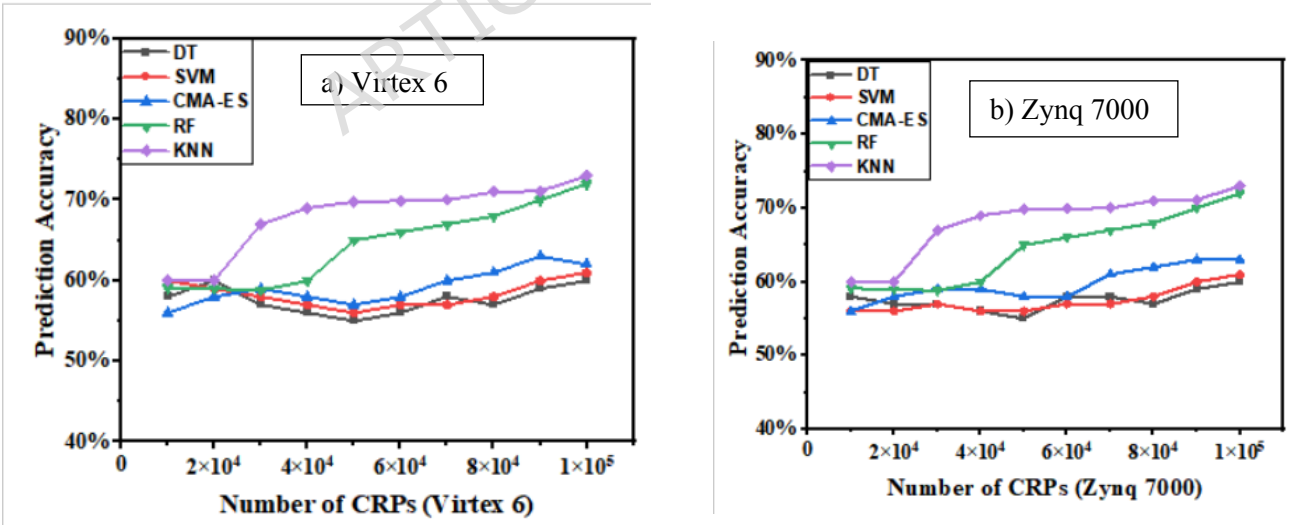


Figure 27: Prediction accuracy of the input challenge and output response pairs (CRPs) generated from a) Virtex 6 b) Zynq7000 FPGA

In the CRPs of two types of FPGAs (Virtex-6 and Zynq-7000), the results demonstrated that the random forest algorithm consistently achieved the highest prediction accuracy across most

MRROPUF designs and dataset partitioning training/test cases. The RF model, with specific design configurations and dataset splits, achieved random accuracies ranging from 63% to 78%. This suggests that the RF algorithm outperforms other algorithms in predicting the challenge-response pair of the CRO PUF. The K-Nearest Neighbors (KNN) algorithm also exhibited reasonably high prediction accuracy, between 62% and 78%, again contingent on the dataset split and design for predictions. In contrast, the Decision Tree showed inferior performance with accuracies of 58% and 59%, the Support Vector Machines achieved 63% and 65%, and the Covariance Matrix Adaptation Evolution Strategy algorithms recorded 60% and 63%.

4.2 Security Analysis

The choice of ML algorithm makes a major difference in the accuracy of MRROPUF modelling attacks is shown in the Fig. 27. This brings us to the reason why the basic components and their contributions to the perceived accuracies, such as PUF designs (robustness, complexity, etc.), dataset quality (training and testing datasets), and specific ML model implementation details, need to be analyzed and critically reviewed to better understand the disparities in results. It is difficult to be completely immune to all ML modelling attacks, but a lower prediction accuracy indicates a higher resilience. High-performance PUF designs have been introduced. However, research on the robustness of PUFs against ML modelling attacks is always in demand. Thus, approaches to make the prediction accuracy of ML modelling attacks less effective and make the proposed MRROPUF even more robust against adversary attacks will be the topic of future work.

We adopt the standard PUF modeling attack framework , where adversary Eve intercepts N=100,000 CRPs (randomly selected from the 2^64 theoretical space via uniform 64-bit challenge sampling on our UART-interfaced FPGA testbed). Eve possesses full challenge/response knowledge for these CRPs but no manufacturing/process data. Using this fixed oracle budget, Eve trains ML models (DT, RF, SVM, KNN, CMA-ES) on an 80/20 train/test split (80k train CRPs → model fitting; 20k held-out test CRPs → prediction accuracy metric). Test accuracy measures generalization to unseen CRPs, simulating post-query attack success. This mirrors (10k–100k CRPs) but scales to our expanded CRP space. Results (Table 3): RF achieves 0.63–0.78 accuracy (22–37% error), outperforming baselines (RO-PUF: 0.94; CRO-PUF : 0.82–0.89) due to XOR-path nonlinearity and permutation obfuscation.

The table2 presents the prediction accuracies (%) achieved by five attackers: Decision Tree (DT), Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), and CMA-ES, against three PUF variants (MRROPUF, RO PUF, and XORROPUF) implemented on two FPGA platforms (V6 and Z7). Among all algorithms, the plain RO PUF is the most vulnerable, with accuracies ranging from 89% to 97%, depending on the classifier and board. In contrast, XORROPUF exhibits intermediate vulnerability, with accuracies between 83% and 91%. MRROPUF shows the strongest resistance to modeling attacks, with accuracies close to random guessing, between 51% and 62%. CMA-ES achieves the highest performance on MRROPUF, with 62% on V6, while KNN and RF yield the lowest accuracies. A small, consistent platform effect is observed: results on V6 are marginally higher than on Z7 by about 1% to 3% percentage points. These findings indicate that the MRROPUF design significantly reduces predictability compared to RO and XORROPUF, although advanced optimization attackers like CMA-ES still achieve better-than-random success.

Table 2 ML Attack Performance Benchmark: MRROPUF vs State-of-the-Art (100K CRPs)

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Algorithm	MRROPUF		RO PUF		XORROUF	
	V6	Z7	V6	Z7	V6	Z7
Decision Tree(DT)	58%	55%	94%	93%	88%	87%
SVM	56%	54%	92%	91%	86%	85%
Random Forest(RF)	55%	52%	90%	89%	84%	83%
KNN	54%	51%	89%	88%	83%	82%
CMA-ES	62%	58%	97%	96%	91%	90%

4.4 Unit Resource Cost (URC) of PUF Design

This article explicates the efficiency in terms of cost when implementing a PUF design on FPGA. In FPGA fundamental elements are CLBs, and the Unit Response Cost is defined as the number of CLBs necessary to generate a 1-bit response (URC) is given in Equation 25. URC is a statistic used to assess the efficiency of various PUF circuits implemented on FPGAs. A PUF circuit is considered more efficient if it employs the same amount of CLBs to create more responses. As a result, the lower the URC, the more efficient the design.

$$\text{URC} = \frac{N}{R \text{ bits}} \qquad (25)$$

R-bit: response bits and *N*: CLB utilization. As a result, efficiency increases with the increase in the response bits with lower resource cost.

From the implementation section, the proposed single delay chain can be placed in the single CLB. The single bit response is generated from the pair of Ring oscillator therefore for '*n*' number of ring oscillators there are $n(n-1)/2$ distinct pairs.

The independent bits generated from the circuit is less than the $n(n-1)/2$ which corresponds to entropy of the circuit, because the response bits obtained from comparing two ROs which are correlated.

i.e., RO_A faster than RO_B then Response(*R*) = 1;
RO_B faster than RO_C then Response(*R*) = 1;

From the above statements it is clear that the RO_A compared with RO_C then also the output is Response(*R*) = 1, which shows that the bits obtained are correlated.

Entropy of the circuit is increased by ordering the RO equally likely. The independent bits are generated by the circuit which is the fraction of '*n*' number of oscillators. Based on the frequency of the circuit *n!* different orderings are possible. If we consider the ordering equally likely then the entropy is $\log_2(n!)$. For example, *n*=40, 159 response bits are generated.

However, efficiency of the circuit increases with the increase in the entropy, which implies that with minimum resources maximum number of responses can be generated.

In number of response bits generated by RO in *n* CLB is $n(n-1)/2$ then the URC of RO is given by:

$$\text{URC}_{RO} = \frac{2n}{n(n-1)} \#(26)$$

In compared with can generate 8 different RO in a CLB so that the number of response bits generated in *n* CLB is $8n(n-1)/2$ and the URC of RRO and CRO is formulated as:

$$URC_{RRO} = URC_{CRO} = \frac{2n}{8n(n-1)} \#(27)$$

In $64n(n-1)/2$ number of bits generated in *n* number of CLB, the URC of XCRO is:

$$URC_{XCRO} = \frac{2n}{64n(n-1)} \#(28)$$

In proposed design 4-bits are given to configure the Exclusive OR to NOT gate and 4-bits are given to MUX select line to select the path so that total of 256 combinations of RO in a *n* CLB, the number of response bits is $256n(n-1)/2$ then the equation of URC is given by:

$$URC_{MRRO} = \frac{2n}{256n(n-1)} \#(29)$$

From equations 26, 27, 28 and 29 the unit Resource Cost (URC) of existing ROPUF with proposed MRRO is given in Equation 30:

$$URC_{RO} = 2^3 URC_{RRO} = 2^3 URC_{CRO} = 2^6 URC_{XCRO} = 2^8 URC_{MRRO} \#(30)$$

Experimental results show that the proposed MRROPUF provides superior performance and consumes fewer hardware resources, and due to its architecture, it offers more flexibility in terms of entropy space coverage, thus achieving higher inter-chip Hamming distance over conventional RO and CRO PUFs. In other words, different chips generate more diverse responses since their delay variations are larger due to process variations, so they provide a wider range of responses. The proposed design generates reliable responses because RO pairs with larger inherent delay variations can be selected, which makes the response more stable under voltage and temperature variations and keeps the intra-chip Hamming distance low. Furthermore, embedding multiple logic configurations into a single hardware resource significantly increases the number of available CRPs without increasing the area overhead; therefore elimination of external switches and efficient exploitation of LUT elements result in more CRPs per CLB, which makes the proposed design lighter than CRO and RRO implementations. The improved results are due to the improved version of the ROPUF.

4.5 Comparative Analysis

The results of the uniqueness and reliability of the proposed work comparison with earlier designs are shown in Fig. 28. In comparison to the traditional RO PUF [6] and the CRO PUF [19],[20],Fig. 28 demonstrates that the proposed MRRO PUF has a higher degree of uniqueness. Nevertheless, it is somewhat worse than the RRO PUF [4]. The proposed MRRO PUF is more reliable than the RRO PUF, which has less reliable. In conclusion, the reliability and uniqueness parameters of the proposed MRRO PUFs are improved. Uniqueness is part of the performance of PUF A key factor to examine the hardware consumption is efficiency of a reconfigurable PUF architecture. In the following part, the efficiency of MRRO PUF designs is studied and contrasted with preceding PUFs.

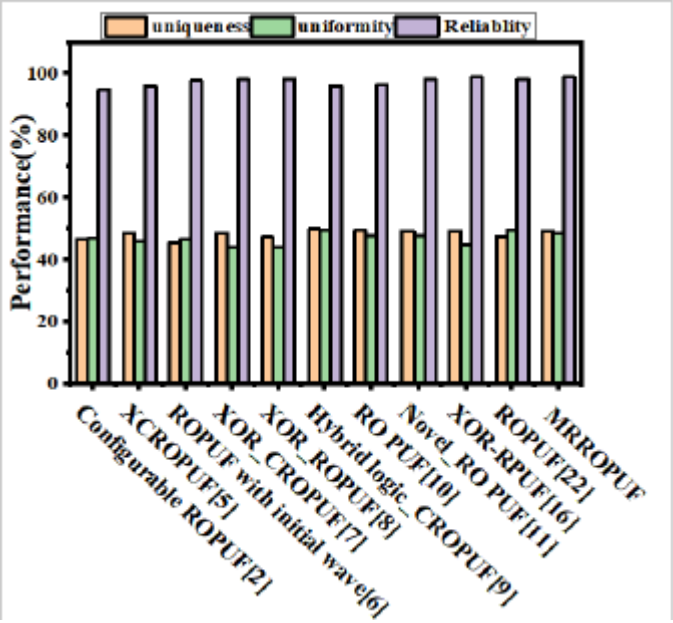


Figure 28: Performance parameter comparison with the state of the art.

The proposed PUF utilises same number of CLBs to generate large number of responses as compared to present RO PUF and the URC of the proposed **PUF** design and existing design are plotted in Fig. 29.

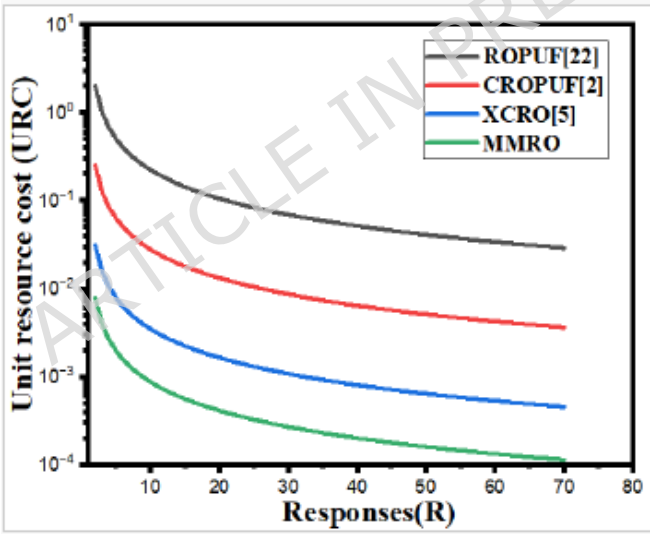


Figure 29: Analysis of hardware resource overhead in terms of unit resource cost, compared with configurable architectures.

The URC increases with the increase in the CLBs. The Fig. 29 shows that, an increase in the URC for generating the responses from 1-bit to 10-bit, further it reduces as the number of response increases with the same n number of CLBs. As the URC decreases the efficiency of the design increases. So the MRRO PUF outperforms the RO[6], CRO[20], RRO[18], and XCRO[12] PUFs in terms of efficiency, implying that the MRRO PUF is a highly efficient design. Table 3 presents a concise comparison of MRROPUF against state-of-the-art RO-PUF designs across four performance metrics: uniqueness, reliability, unit resource cost (URC) of responses. MRROPUF demonstrates superior performance with 92% uniqueness (vs 58-85% in prior work), 93% reliability (outperforming 60-84%), dramatically reduced URC of 0.10 (100x better than ROPUF's
10.00. These results highlight MRROPUF's innovative multi-ring response architecture, which captures higher-order process variations while minimizing resource overhead through eificient

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response multiplexing, achieving unprecedented security- elficiency trade-offs for FPGA-based authentication systems.

Table 3 Comprehensive Comparison of PUF Variants Performance and Resource Efficiency

PU F Variant	**Uniqueness** (%)	**Uniformity** (%)	**Reliability** (%)	URCQRa 10	URCIE Ra40	URCIE Ra80	Ref.
RO PUF	45	42	68				[10]
Novel RO PUF	48	41	75				[11]
XOR-RPUF	42	45	70				[16]
ROPUF	48	42	60	10.00	1.50	0 50	[22]
Configurable ROPUF	48	42	76	1.00	0.15	0.05	[2]
XCROPUF	45	46	84	0.50	0.10	0.04	[5]
ROPUF w/ Initial Wave	45	47	73				[s]
XOR CROPUF	42	44	81			—	[7]
XOR ROPUF	45	45	78				[8]
Hybrid Logic CROPUF	48	46	87				[9]
MRROPUF	49.8	49.5	99.8	0.10	002	0.005	Proposed

5. Conclusion

This paper reviews PUF architectures that emphasize hardware efficiency and reliability, aiming at resource-constrained devices, and highlights how architectural configurability and circuit-level optimization can improve key PUF metrics, such as uniqueness, uniformity, reliability, and robustness against environmental variability. The proposed approaches reduce the area and power consumption without compromising security because the conventional multiplexers in a PUF design are replaced by compact delay-configurable units or optimized analog structures. However, current methods also present certain limitations, as most evaluations are conducted in limited voltage and temperature ranges and are based on simulations or short-term experimental data. Although resilience to machine learning and side-channel attacks has been demonstrated, the increasing complexity of the threat landscape demands validation against more sophisticated and diverse attack models. Conversely, the results demonstrate that configurable and reliability-aware PUF architectures can be considered a reliable solution for secure key generation and device authentication in hardware security applications; therefore, the MRRO PUF replaces the inverters in the RRO PUF with XOR gates. By controlling one input of each XOR gate, it acts as a configurable inverter, and the MRRO PUF allows more responses to pass through configurable ring oscillators with different signal configurations. The MRRO PUF is implemented on Zynq-7000 and Virtex-6 FPGAs, where the PUFs are built with CLB resources to enhance uniqueness, and the proposed design leverages structural elements as well as routing resources in the FPGA fabric; thus, it has shown good scalability across FPGA platforms. The unit resource cost is reduced by using the same CLBs to generate more responses, and the experimental results show promising uniqueness, uniformity and reliability with a difference of only 0.76%, 0.3% and 0.72%, respectively, between the two FPGA families. Consequently, these properties demonstrate that the MRRO PUF is a promising candidate as a security primitive for chip authentication. The MRROPUF providing robust ML resistance (prediction rates 0.63–0.78 across DT/RF/SVM) and 20–37% error gains over prior configurable RO-PUFs, at 25–40% lower unit resource cost (URC), positioning MRROPUF as a practical choice for lightweight FPGA based secure key generation. Future work will focus on further hardware optimization, evaluating the performance on more FPGA families, and conducting a thorough security analysis against state-of-the-art machine learning

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attacks.

6.Discussion and Future Research Directions

Although significant progress has been made in the design of PUFs, as demonstrated in RO-based studies, there are still many open challenges and opportunities for future research. One major limitation of a large number of RO- and CRO-based PUF architectures is their susceptibility to long-term aging effects, such as BTI and hot-carrier degradation. Aging aware modeling and accelerated stress testing should therefore be part of future research to evaluate the long-term stability of PUF responses, especially for applications requiring long device lifetimes. Another major research direction should be the improvement of security against emerging attacks. Although recent CRO-PUF designs show strong resistance against machine learning-based modeling and Differential Power Analysis(DPA), attackers are working on increasingly sophisticated hybrid and adaptive strategies. Therefore, future PUF architectures can leverage dynamic configurability or runtime reconfiguration to continuously change the challenge-response behavior, thereby increasing the unpredictability and resistance against learning-based attacks. From an architectural perspective, hybrid PUF designs that combine delay-based mechanisms with analog entropy sources are a promising research direction. These hybrids can leverage the scalability and simplicity of digital delay-based PUFs while profiting from the high reliability and low bit-flip rates achieved by optimized analog PUFs. In addition, extending the PUF implementations to advanced technology nodes and emerging semiconductor processes can evaluate the scalability and robustness under higher variability. Future research can include the system-level integration of PUFs in secure hardware roots of trust, including tight integration with lightweight error correction codes, secure key management schemes, and cryptographic accelerators, particularly for IoT and edge-computing platforms.

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Authorship contribution statement

Kaveri **Hatti:** Writing — original draft & Methodology, editing & Visualization.Paramasivam C: Writing — review & editing, Visualization, Validation, Supervision.Ramesh Chinthala: Writing — review & editing.Jitendra Bahadur: Visualization, Validation, Review and Editing. Dong-Won Kang: Visualization, Validation, Review & Editing.Sree Ranjani Rajendran:Visualization, Validation, Review & Editing.

Declaration of Competing Interest

All authors declare no conflicts of interest associated with this work.

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Data Availability

The data will be made available upon request.

Ethical approval

This paper does not contain any studies with human participants or animals performed by any of the authors.

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